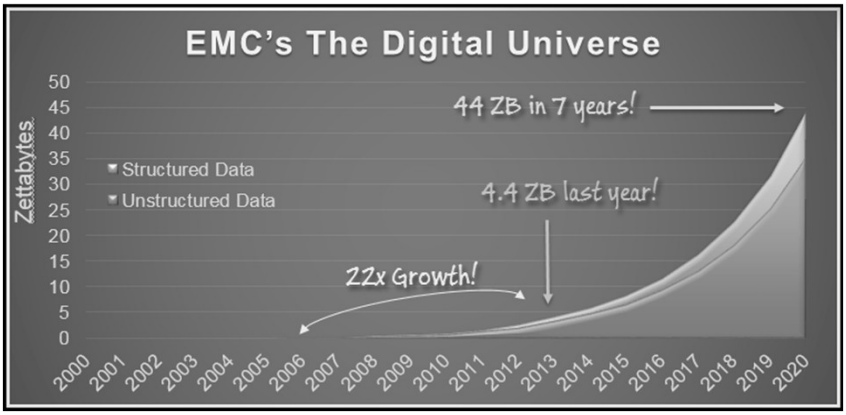
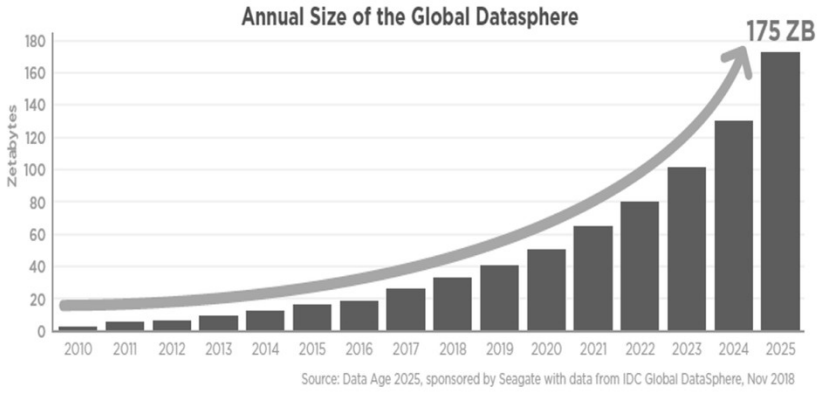
Big data storage and processing - introduction

Vũ Tuyết Trinh
SoICT - HUST

How big is big data?

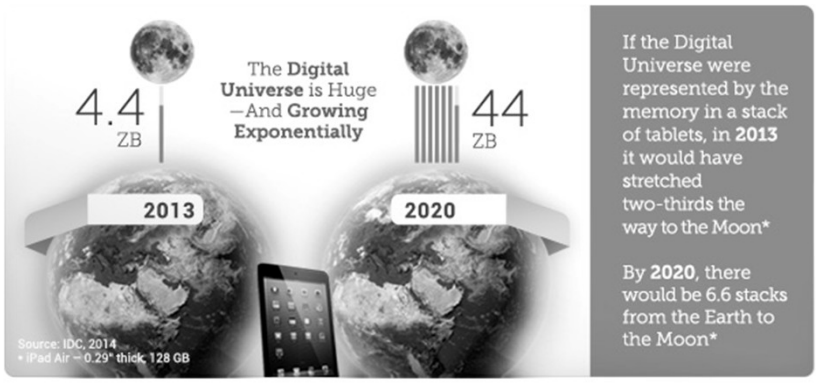




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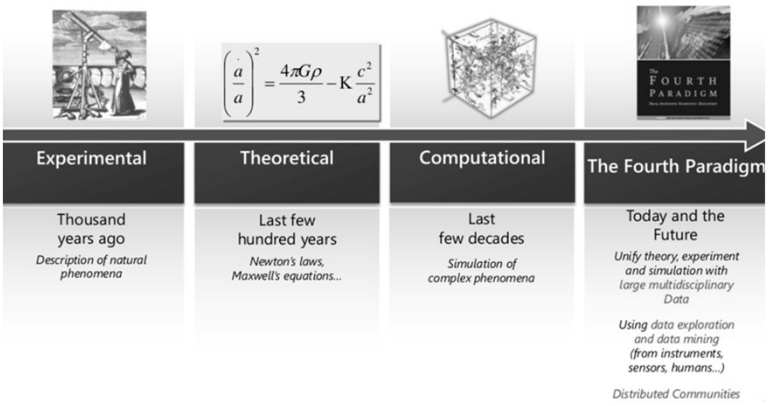
How big is big data?



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Data science: The 4th paradigm for scientific discovery



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Big data in 2008

http://www.wired.com/wired/issue/16-07 **September 2008**



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Big data in 2014



THE AVERAGE PERSON TODAY PROCESSES MORE DATA IN A SINGLE DAY THAN A PERSON IN THE 1500'S DID IN AN ENTIRE LIFETIME

LOOK TO THE LEFT, and you see Times Square at dusk. Look to the right, and you see the same location at midmorning. Internationally acclaimed photographer Stephen Wilkes's time-altering image of New York's Times Square is part of his body of work titled ***Day to Night***.

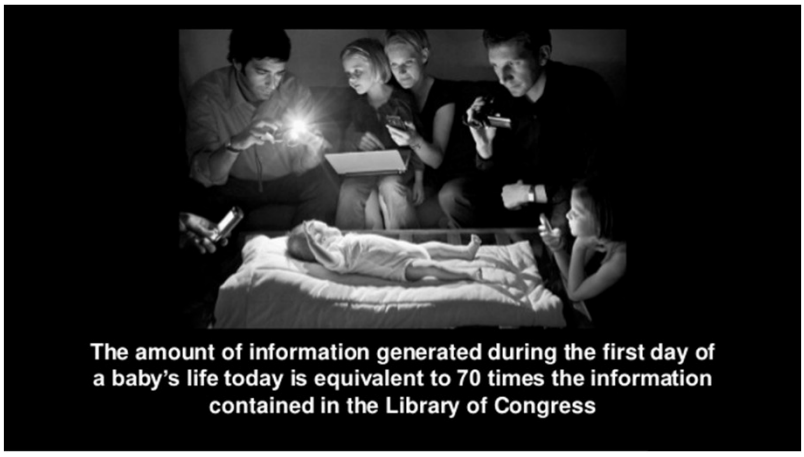
The image was created by blending more than 1,400 separate photos taken over the course of 15 hours—a meticulous process that took him nearly three months.

PHOTO: STEPHEN WILKES

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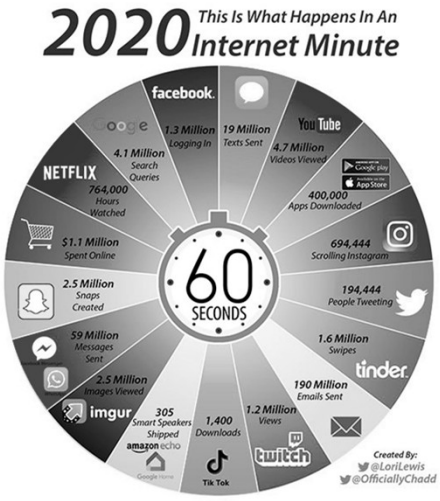
Big data today



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Big numbers

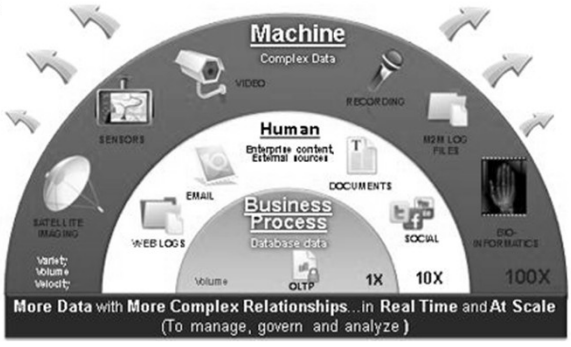


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Big data sources

- E-commerce
- Social networks
- Internet of things
- Data-intensive experiments (bioinformatics, quantum physi



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Data is the new oil

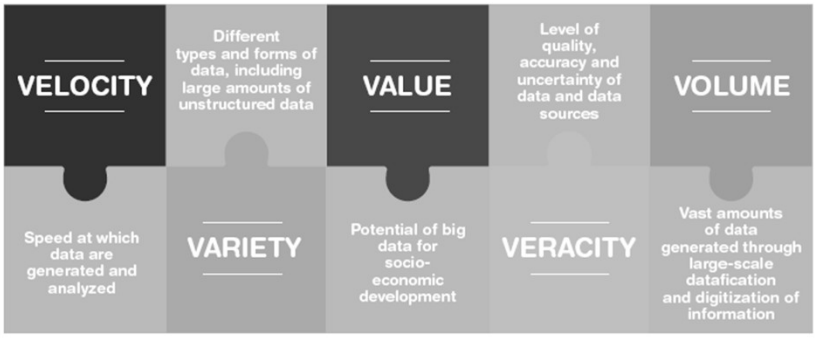




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Big data 5'V

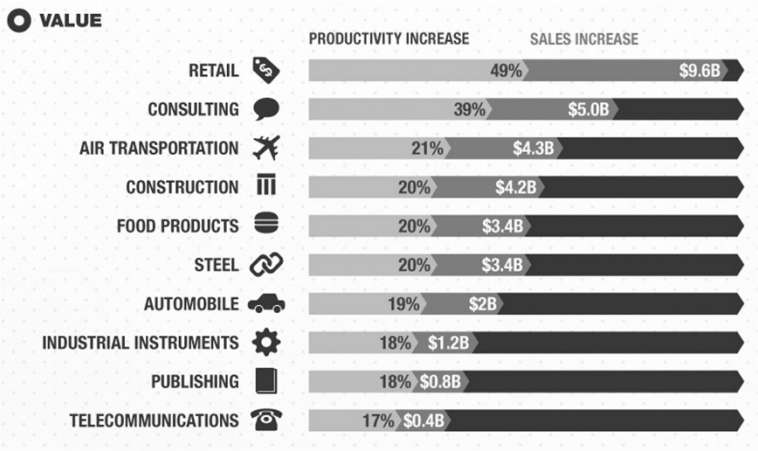


Big data is a term for data sets that are so large or complex that traditional data processing application software is inadequate to deal with them (wikipedia)

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Big data – big value



source: wipro.com

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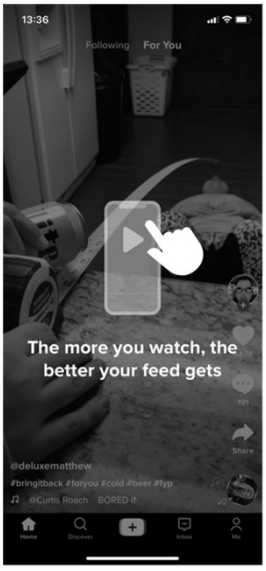
Amazon



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Tiktok

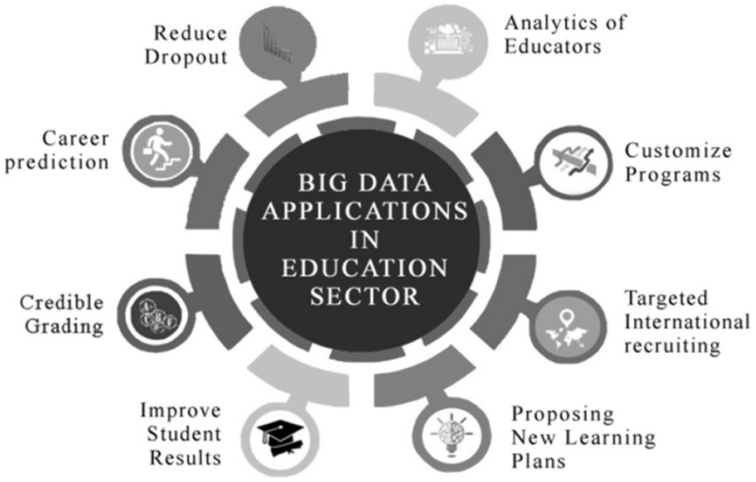
- TikTok has 1.04 billion monthly active users globally as of 2024.
- TikTok users spend 58 minutes and 24 seconds on the app daily as of 2024.
- The majority of TikTok users are between the ages of 18 to 34, at 69.3%.



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Big Data in education industry



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Big Data in healthcare industry

- Reduce costs of treatments, unnecessary diagnosis.
- Predict outbreaks of epidemics and preventive measures.
- Avoid preventable diseases



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Big Data in government sector

- Welfare Schemes
  - Make faster and informed decisions
  - Identify areas that are in immediate need of attention
  - Overcome national challenges such as unemployment, terrorism,.
- Cyber Security
  - deceit recognition.
  - Catching tax evaders.



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Big Data in media and entertainment industry

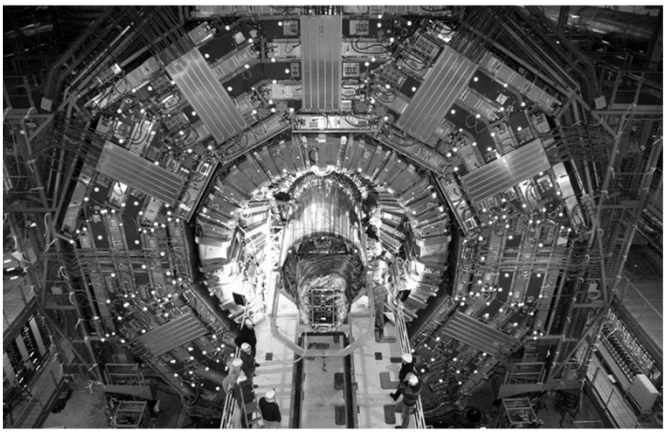
- Predicting the interests of audiences
- Optimized or on-demand scheduling of media streams in digital media distribution platforms
- Getting insights from customer reviews
- Effective targeting of the advertisements
- Example
  - Spotify, Amazon Prime



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Big data in scientific discovery



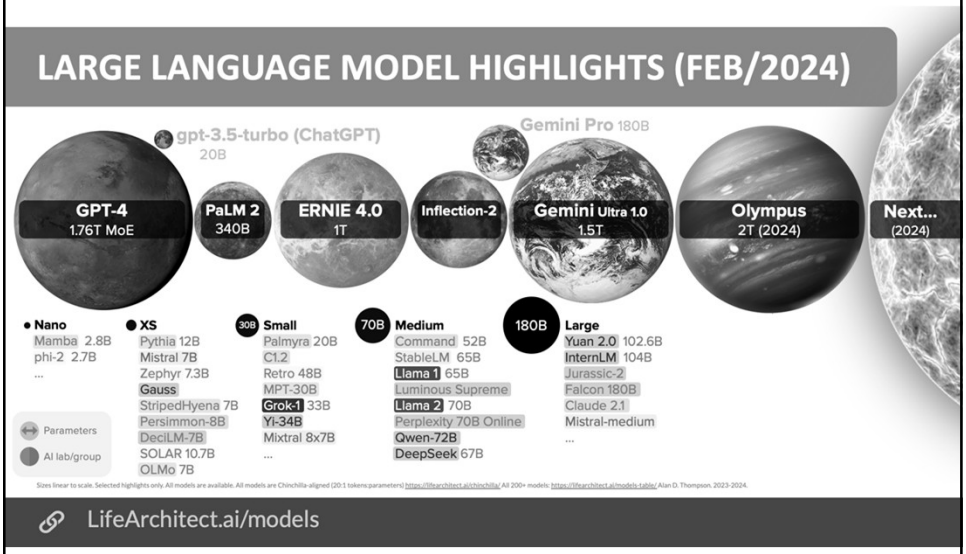
CERN's Large Hydron Collider (LHC) generates 15 PB a year

20

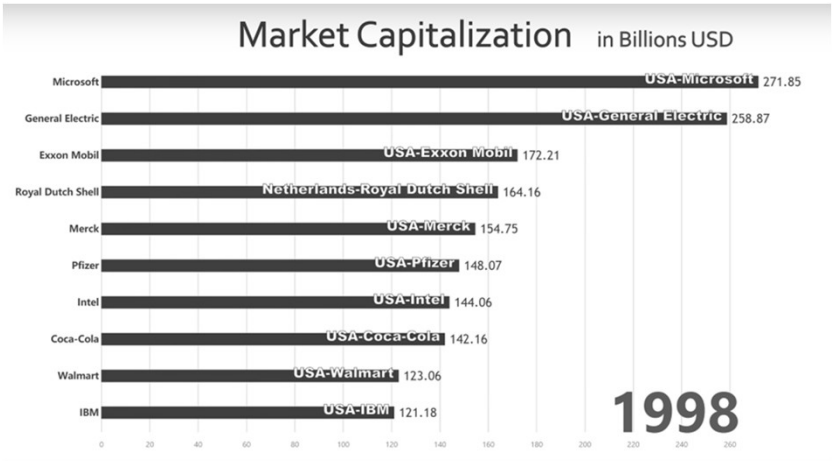
Maximilien Brice, © CERN

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The world is running out of data to train AI

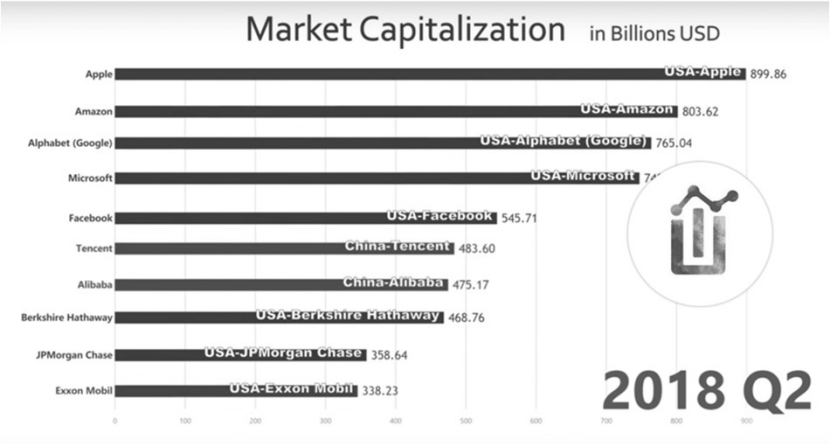


Top 10 Company Market Cap Ranking History (1998-2018)



https://www.youtube.com/watch?v=fobx4wIS6W0

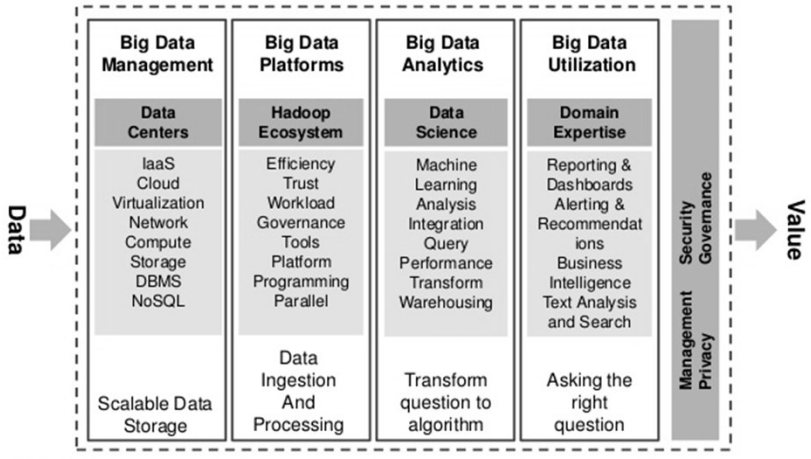
Top 10 Company Market Cap Ranking History (1998-2018)



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Big data technology stack



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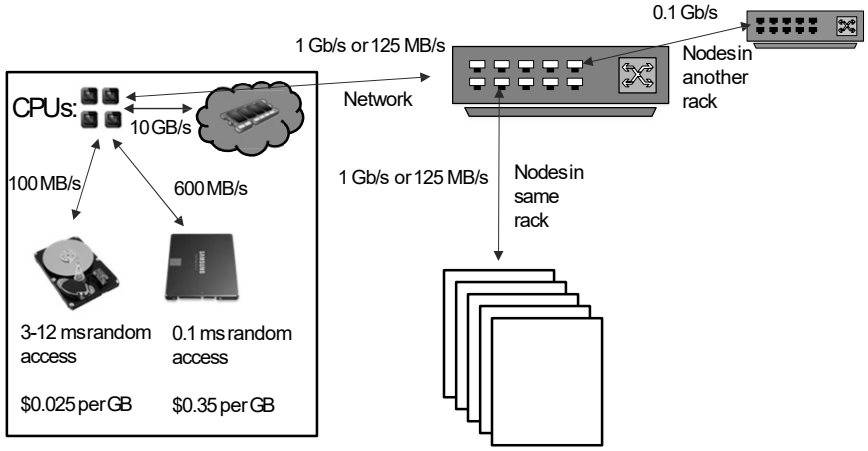
Scalable data management

- Scalability
  - Able to manage incresingly big volume of data
- Accessibility
  - Able to maintain efficiciency in reading and writing data (I/O) into data storage systems
- Transparency
  - In distributed environment, users should be able to access data over the network as easily as if the data were stored locally.
  - Users should not have to know the physical location of data to access it.
- Availability
  - Fault tolerance
  - The number of users, system failures, or other consequences of distribution shouldn't compromise the availability.

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Data I/O landscape



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Scalable data ingestion and processing

- Data ingestion
  - Data from different complementing information systems is **to be combined to gain a more comprehensive basis** to satisfy the need
  - How to ingest data efficiently from various, distributed heterogeneous sources?
    - Different data formats
    - Different data models and schemas
    - Security and privacy
- Data processing
  - How to process massive volume of data in a timely fashion?
  - How to process massive stream of data in a real-time fashion?
  - Traditional parallel, distributed processing (OpenMP, MPI)
    - Big learning curve
    - Scalability is limited
    - Fault tolerence is hard to achive
    - Expensive, high performance computing infrastructure
  - Novel realtime processing architecture
    - Eg. Mini-batch in Spark streaming
    - Eg. Complex event processing in Apache Flink

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Scalable analytic algorithms

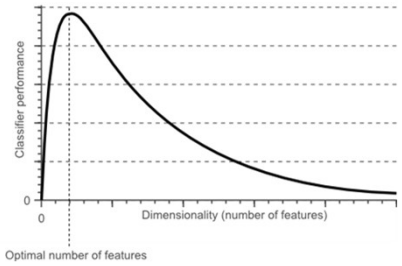
- Challenges
  - Big volume
  - Big dimensionality
  - Realtime processing
- Scaling-up Machine Learning algorithms
  - Adapting the algorithm to handle Big Data in a single machine.
    - Eg. Sub-sampling
    - Eg. Principal component analysis
    - Eg. feature extraction and feature selection
  - Scaling-up algorithms by parallelism
    - Eg. k-nn classification based on MapReduce
    - Eg. scaling-up support vector machines (SVM) by a divide and-conquer approach

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Eg. Curse of dimensionality

- The required number of samples (to achieve the same accuracy) grows exponentionally with the number of variables!
- In practice: number of training examples is fixed!
  - => the classifier's performance usually will degrade for a large number of features!



In fact, after a certain point, increasing the dimensionality of the problem by adding new features would actually degrade the performance of classifier.

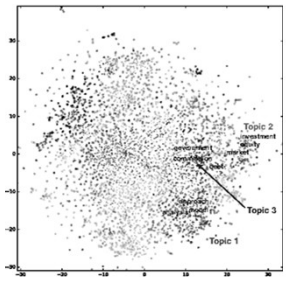
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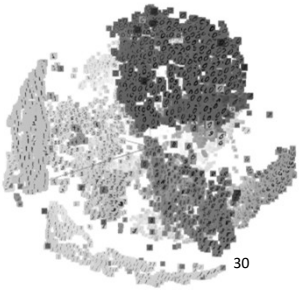
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Utilization and interpretability of big data

- Domain expertise to findout problems and interprete analytics results
- Scalable visualization and interpretability of million data points
  - to facilitate their interpretability and understanding





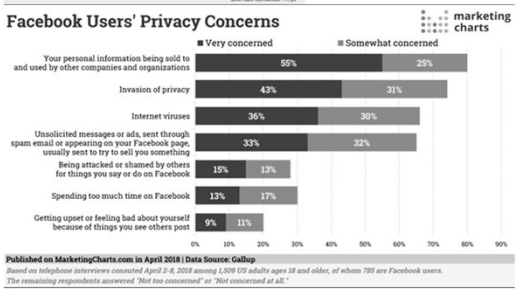


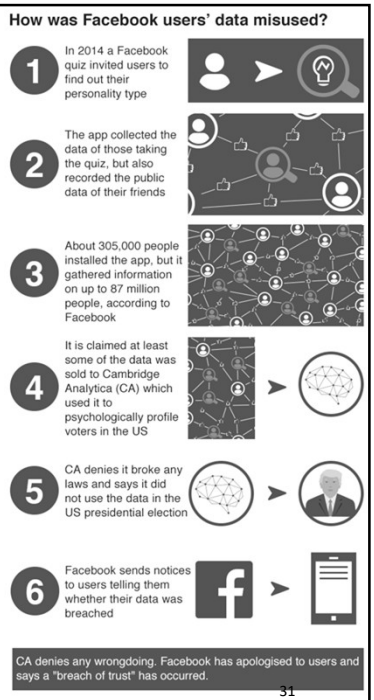
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Privacy and security







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Big data job trends



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Talent shortage in big data

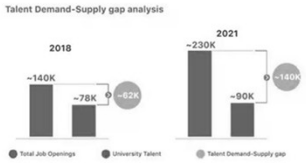


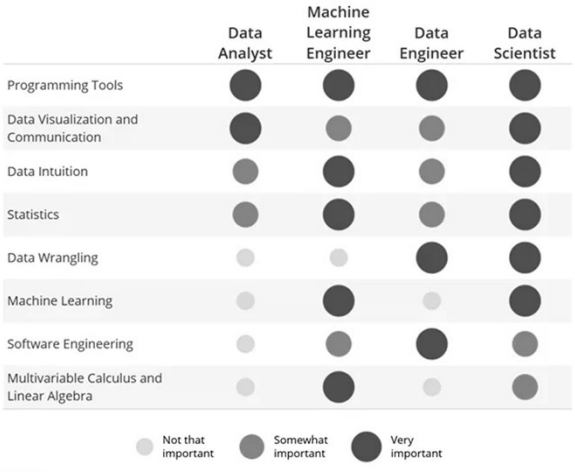
Table 2. Summary Demand Statistics

DSA Framework Category	Number of Postings in 2015	Projected 5-Year Growth	Estimated Postings for 2020	Average Time to Fill (Days)	Average Annual Salary
All	2,352,681	15%	2,716,425	45	$80,265
Data-Driven Decision Makers	812,099	14%	922,428	48	$91,467
Functional Analysts	770,441	17%	901,743	40	$69,162
Data Systems Developers	558,326	15%	641,635	50	$78,553
Data Analysts	124,325	16%	143,926	38	$69,949
Data Scientists & Advanced Analysts	48,347	28%	61,799	46	$94,576
Analytics Managers	39,143	15%	44,894	43	$105,909

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Big data skill set

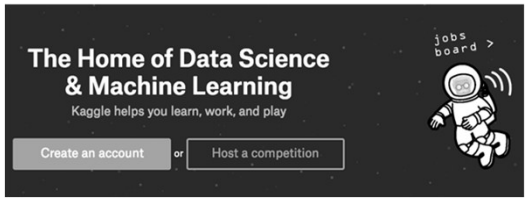


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How to land big data related jobs

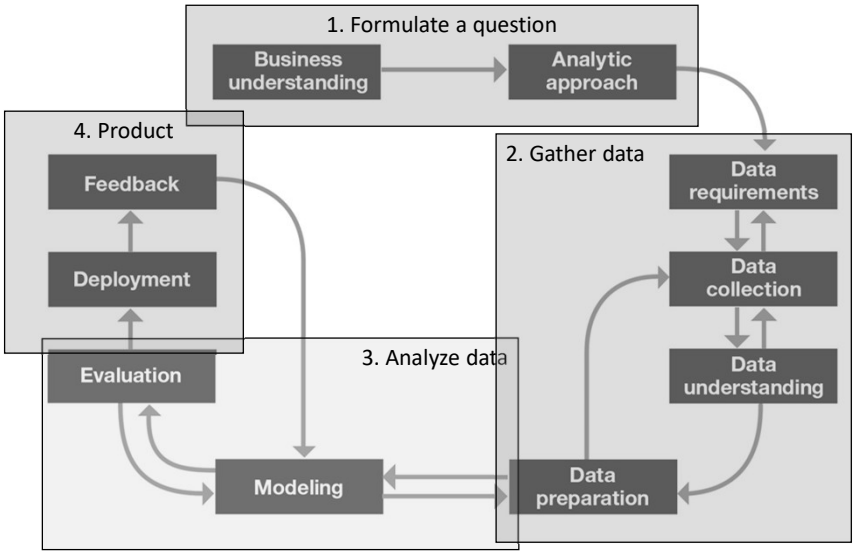
- Learn to code
  - Coursera
  - Udacity
  - Freecodecamp
  - Codecademy
- Math, Stats and machine learning
  - Kaggle
- Hadoop, NoSQL, Spark
- Visualization and Reporting
  - Tableau
  - Pentahoo
- Meetup & Share
- Find a mentor
- Internships, projects



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Data science method

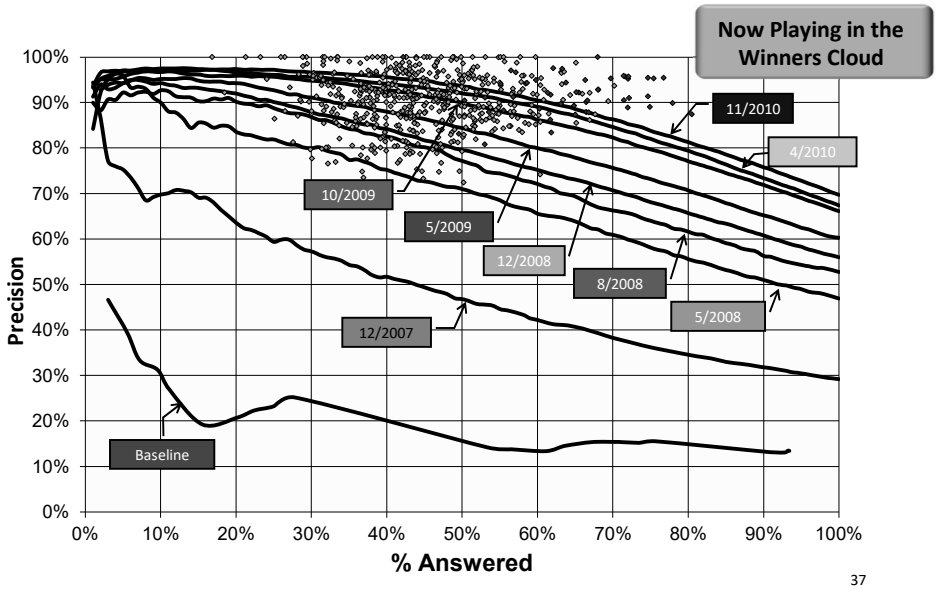


Source: *Foundational Methodology for Data Science*, IBM, 2015

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DeepQA: Incremental Progress in Precision and Confidence 6/2007-11/2010

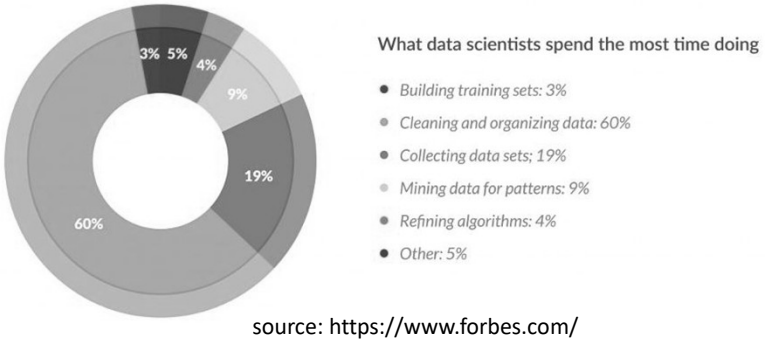


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Cleaning big data: most time-consuming, least enjoyable data science task

- Data preparation accounts for about 80% of the work of data scientists



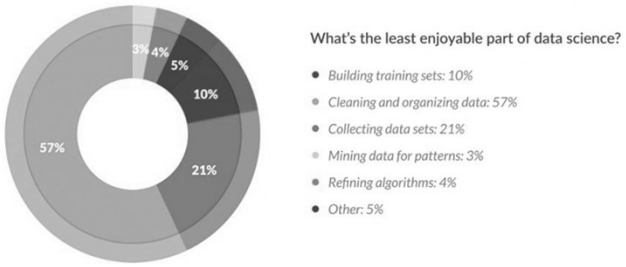
source: https://www.forbes.com/

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Cleaning big data: most time-consuming, least enjoyable data science task

- 57% of data scientists regard cleaning and organizing data as the least enjoyable part of their work and 19% say this about collecting data sets.



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References

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